Lecture 6
Ceramic Materials for Various Applications

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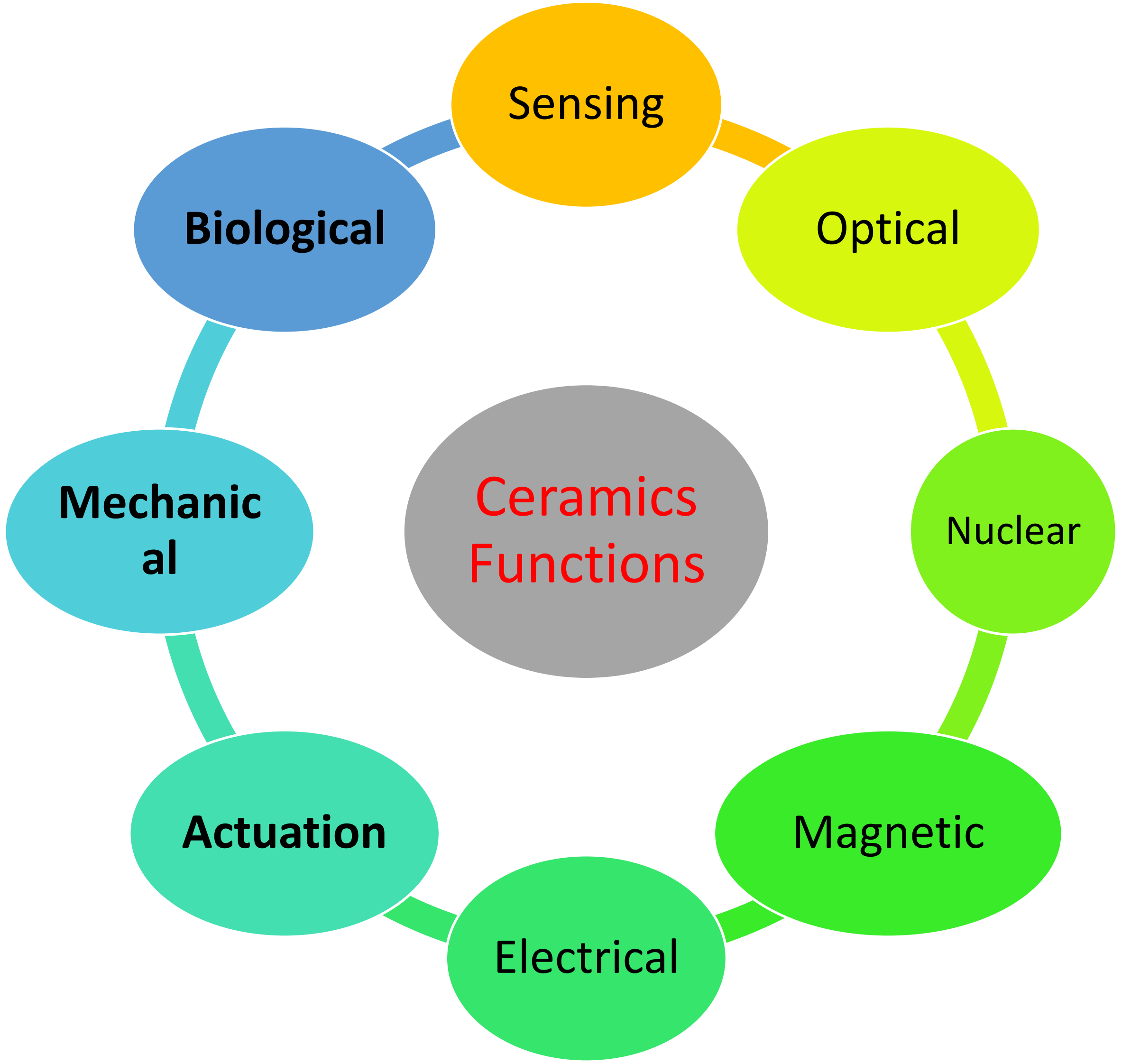
APPLICATIONS OF CERAMICS

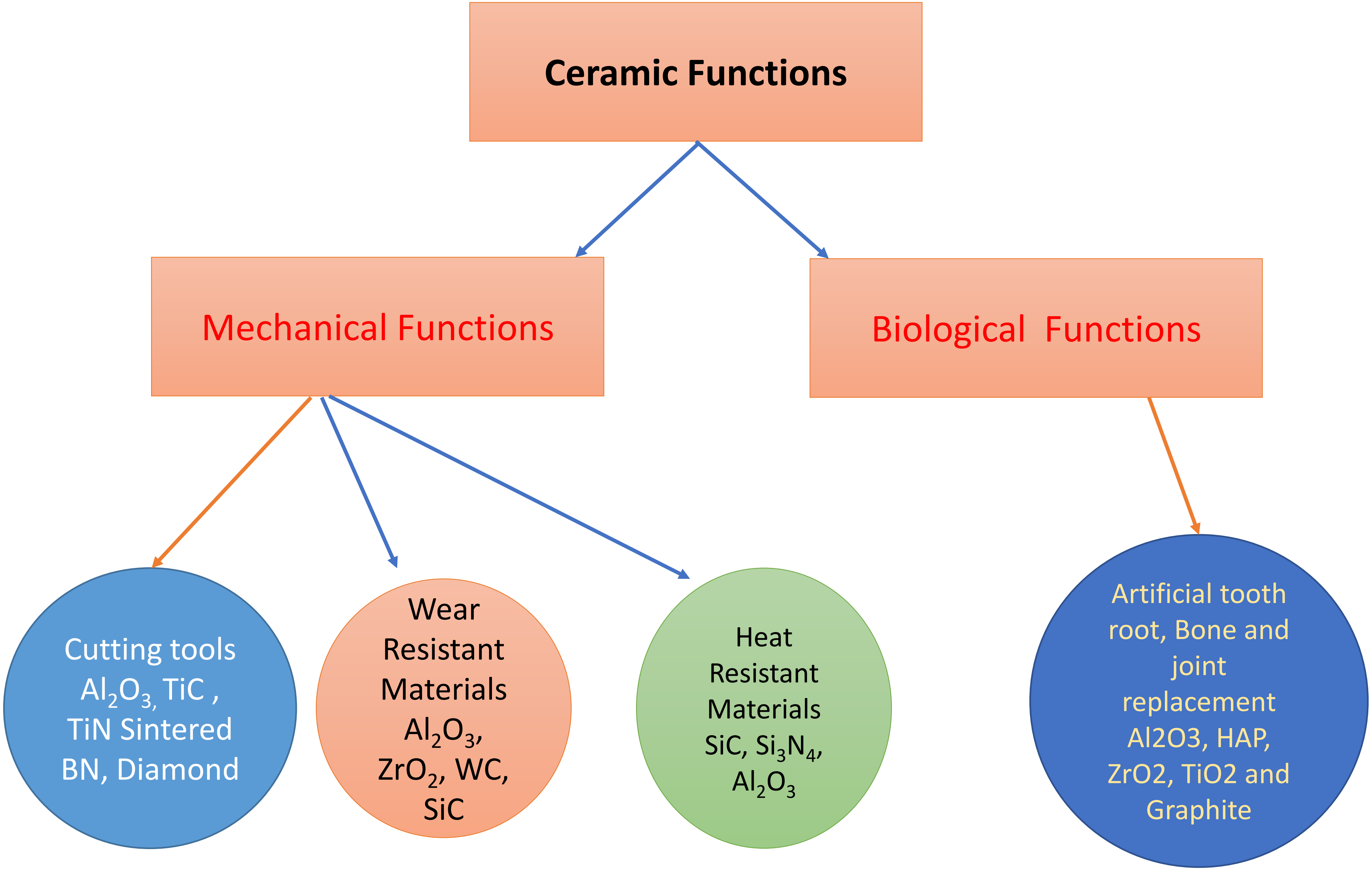
Ceramic materials are used extensively in manufacturing and perform a variety of functions

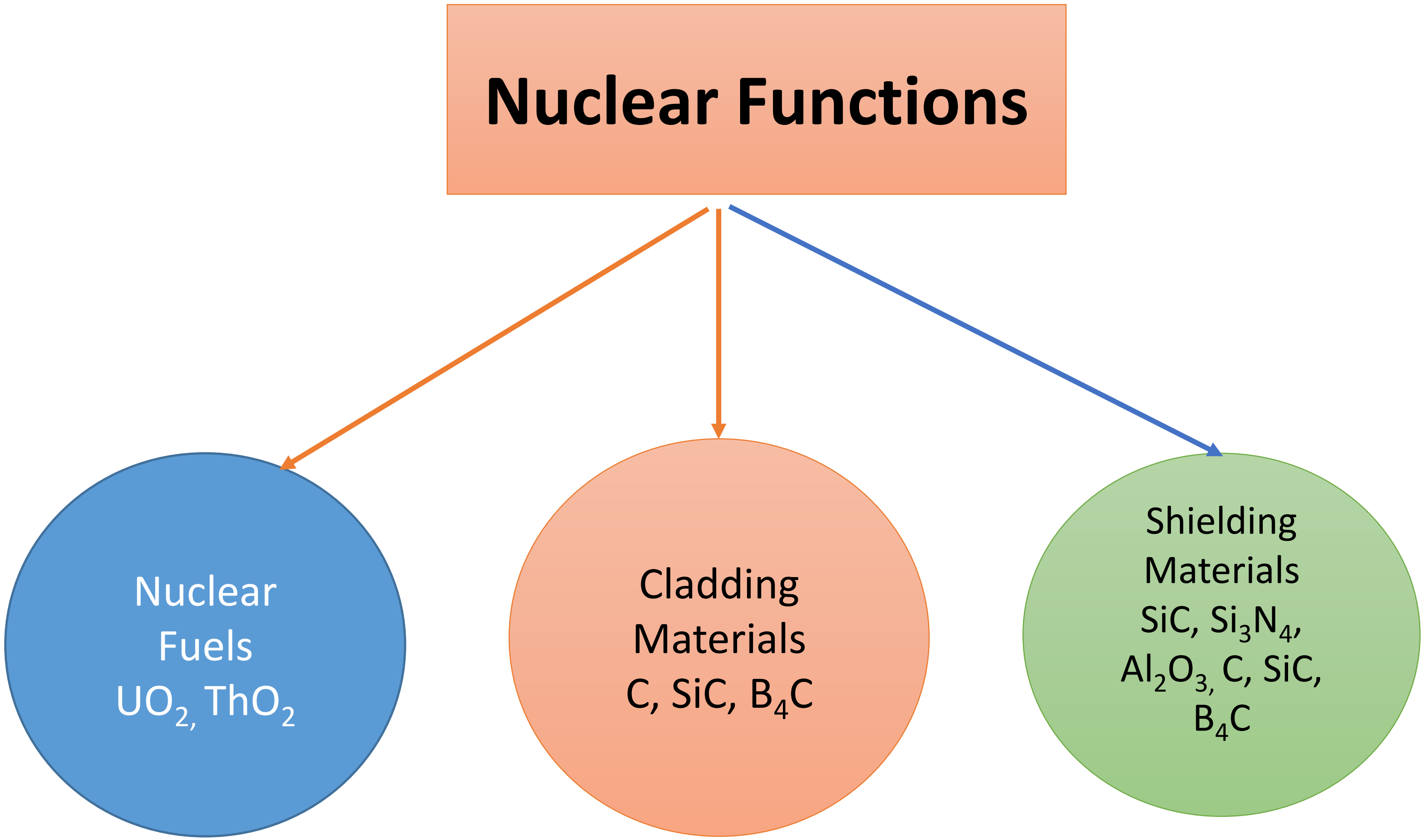
Broad category of functions includes:

- **Heat resistance**
- **Wear resistance**
- **Chemical Resistance**
- **Metal Forming & Finishing**

Ceramic functions




Ceramic Functions
Mechanical Functions
Biological Functions
Cutting tools Al_2O_3, TiC , TiN Sintered BN, Diamond
Wear Resistant Materials Al_2O_3, ZrO_2, WC, SiC
Heat Resistant Materials SiC, Si_3N_4, Al_2O_3
Artificial tooth root, Bone and joint replacement Al2O3, HAP, ZrO2, TiO2 and Graphite


Nuclear Functions
Nuclear Fuels
UO_2, ThO_2
Cladding Materials
C, SiC, B_4C
Shielding Materials
SiC, Si_3N_4, Al_2O_3, C, SiC, B_4C

Ceramics functions as heat resistance

Heat resistant ceramics

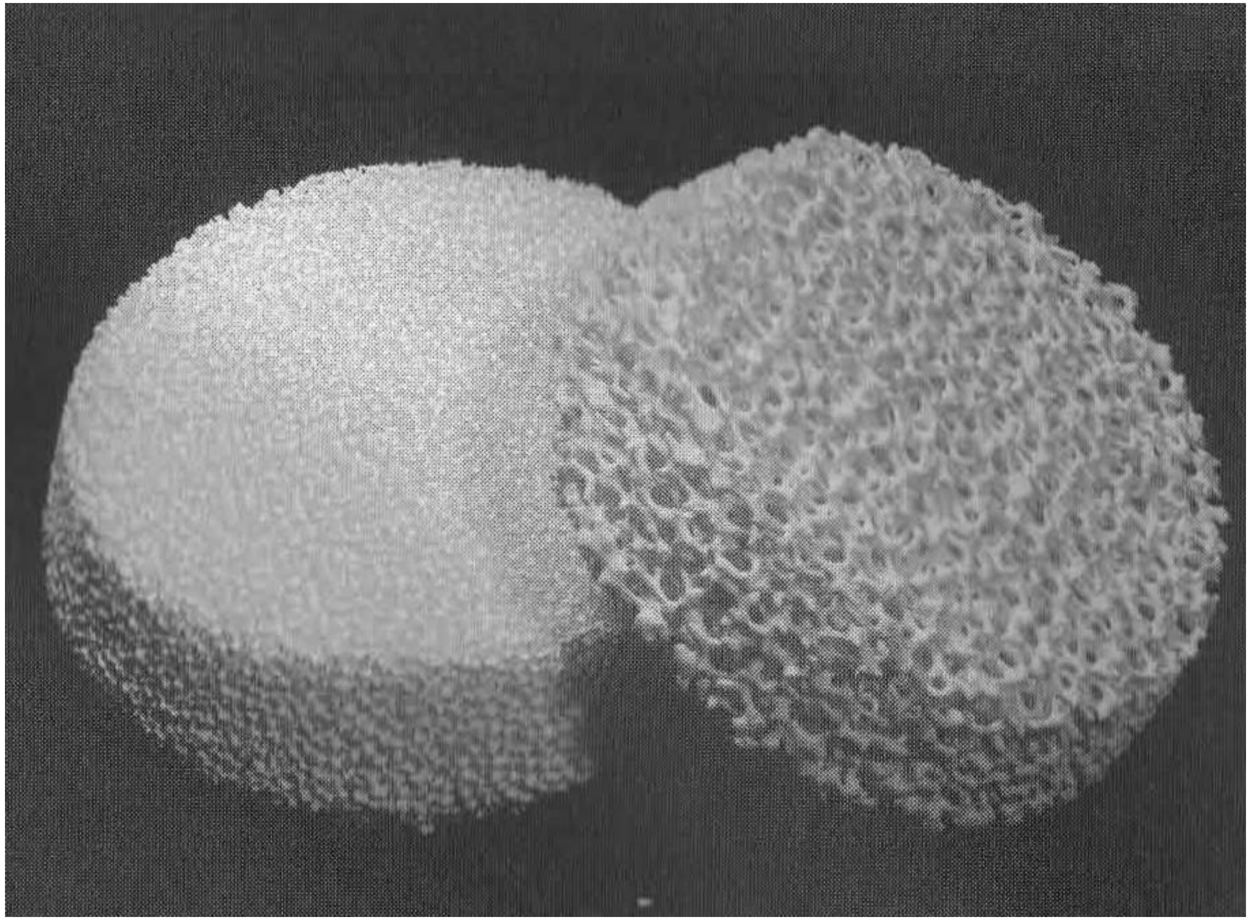
- **SiC, Si_3N_4 and Al_2O_3**

Find applications in fabrication of metal products such as investment casting, extrusion product, wire, etc.

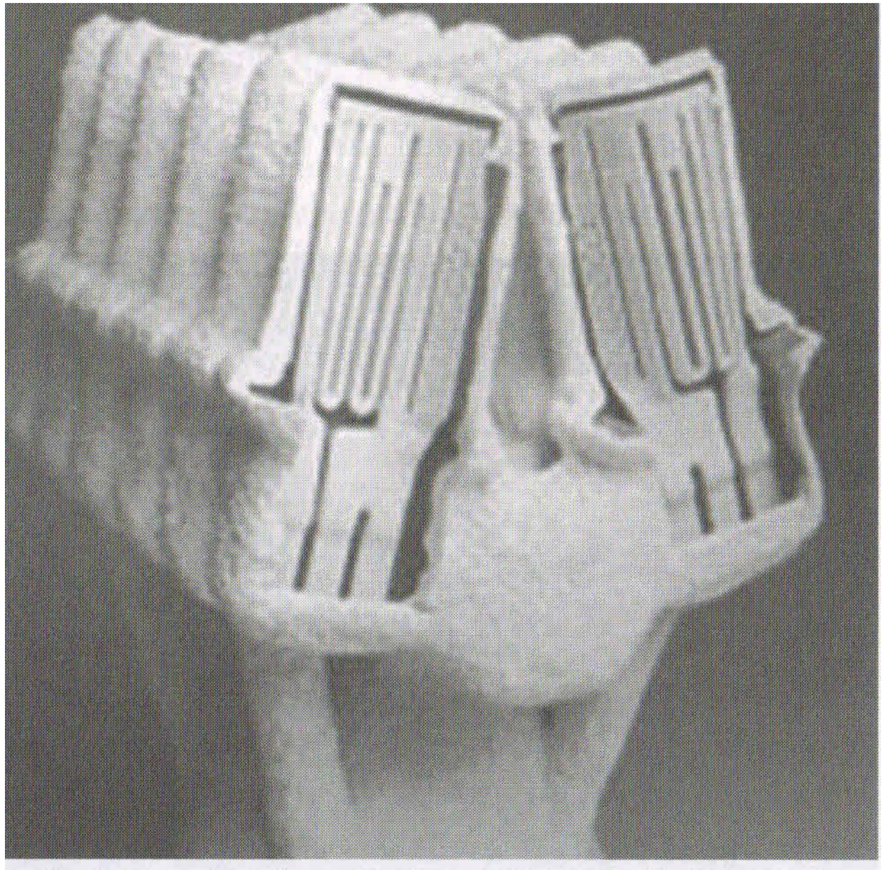
- **Crucibles, moulds, furnace lining and often even the heating elements used in melting, refining and net-shape forming of metals, liners in hot extrusion press, wire-drawing die, etc. are made of ceramics.**

- **Other high-temperature manufacturing applications of ceramics include heat exchangers, welding (nozzles, caps, and quench blocks), and braze fixtures.**

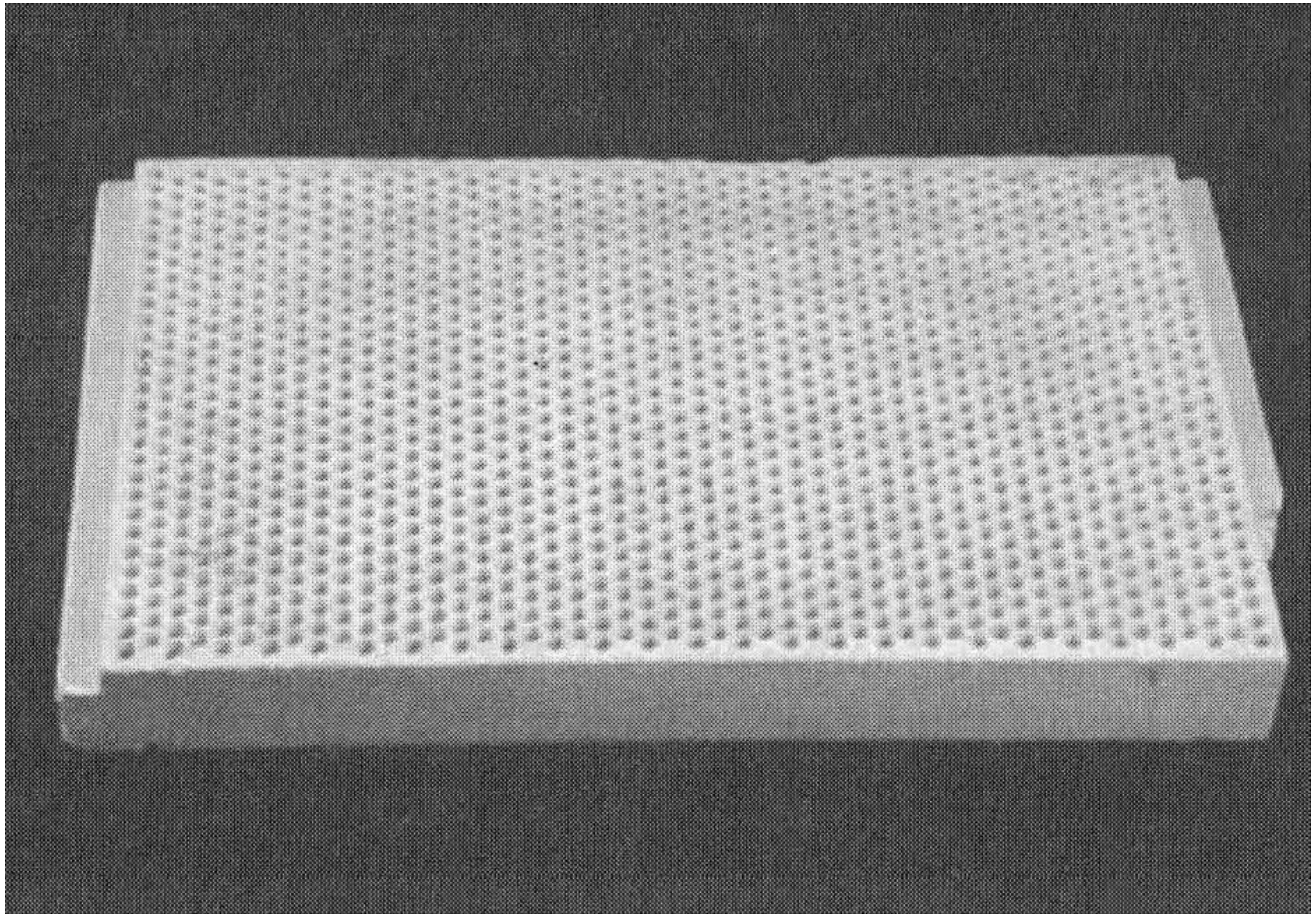
Zirconia crucibles



Ceramic filters for screening debris from molten metals during casting and thus greatly reducing the % of reject metal parts.



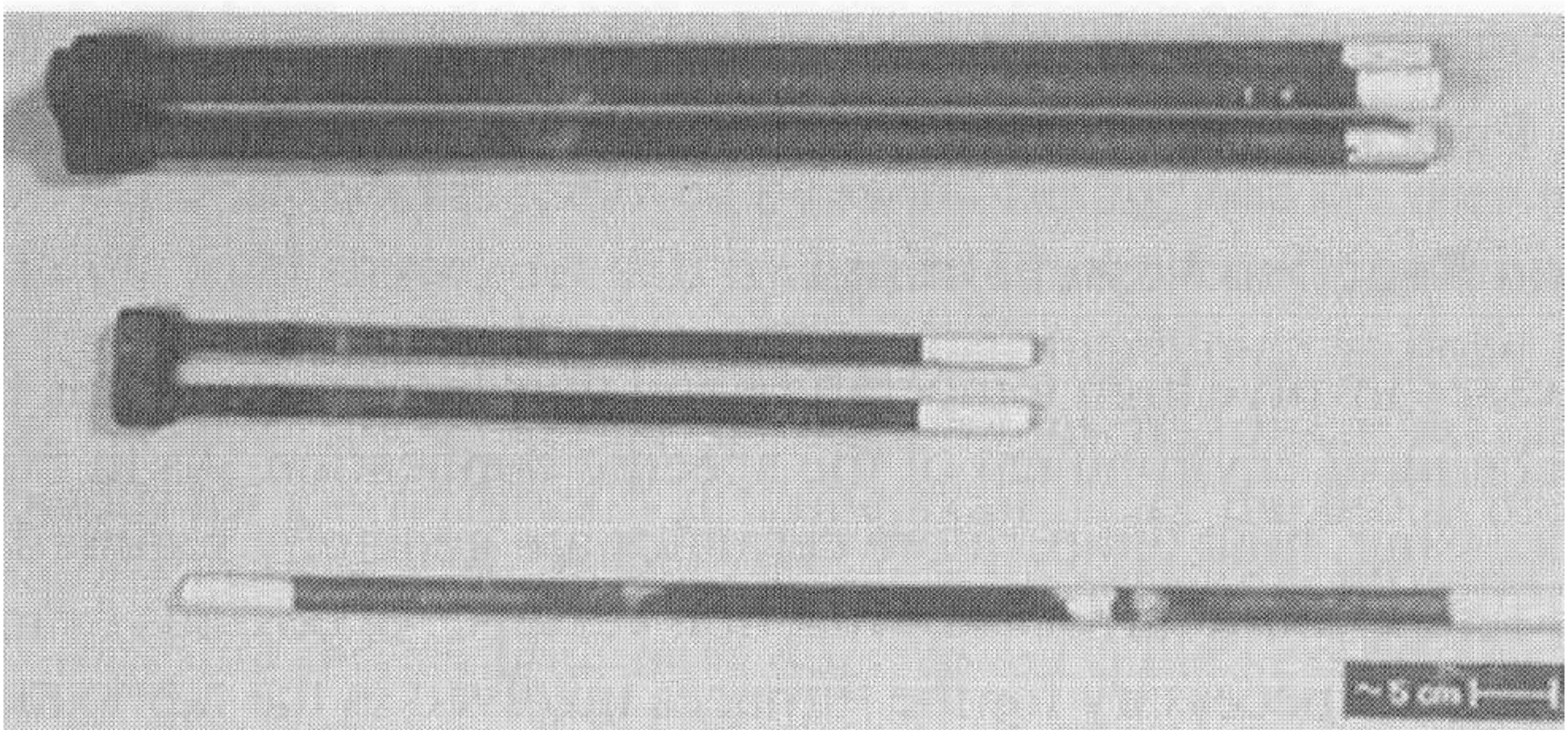
Cutaway view of a ceramic investment casting mould for fabrication of gas turbine engine rotor blades showing the ceramic cores for producing the complex internal cooling passages in the cast metal blades



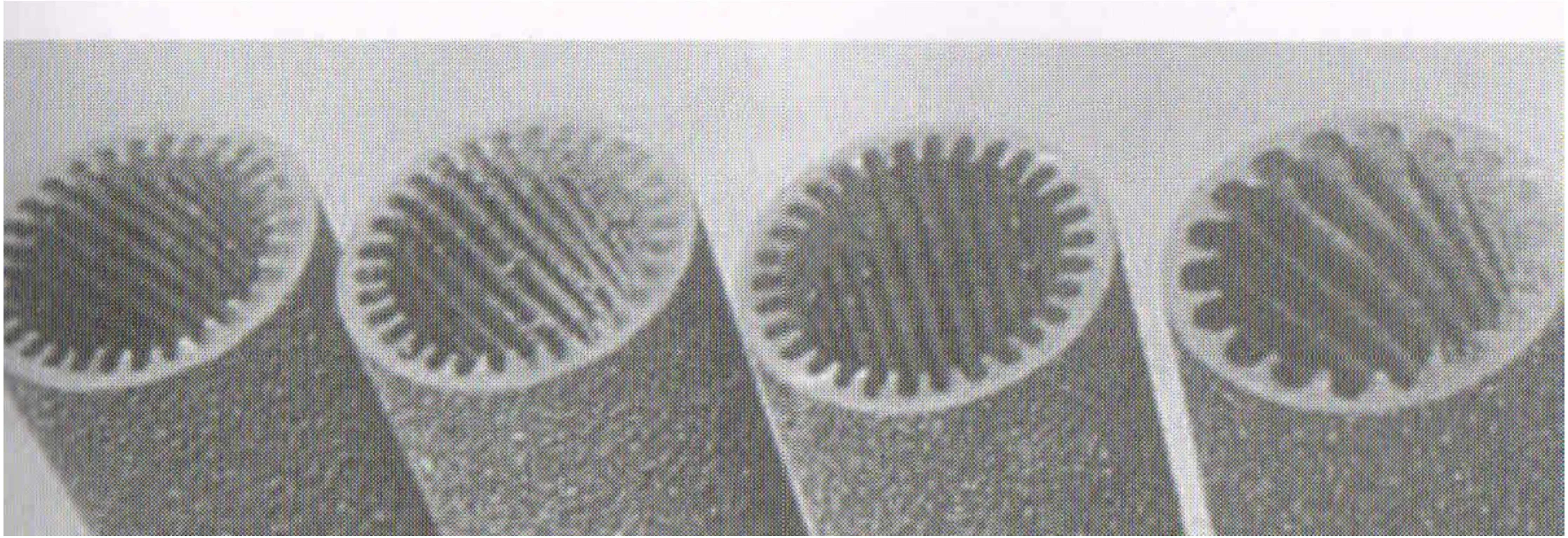
Perforated ceramic plate for a low-pollution, radiant surface burner. Fuel and air mixed and forced through the array of holes such that combustion occurs on the surface of the plate. This produces uniform and efficient heating with low in nitrogen oxide (NO_x) pollutants.

Perforated ceramic plate surface burners

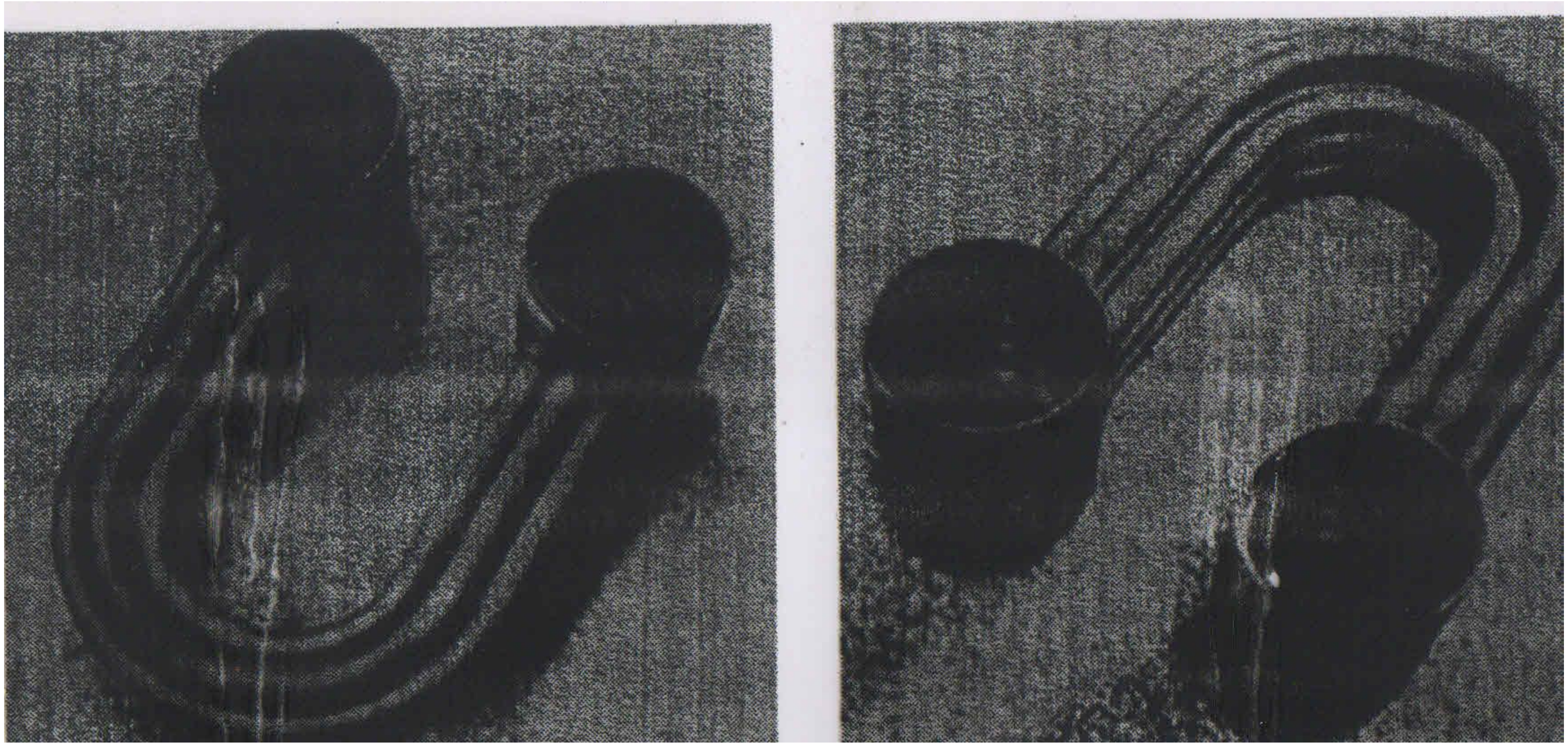
These surface burners have been used in applications, such as boilers, H.T. furnaces, glass windshield shaping and also for cooking Pizza.



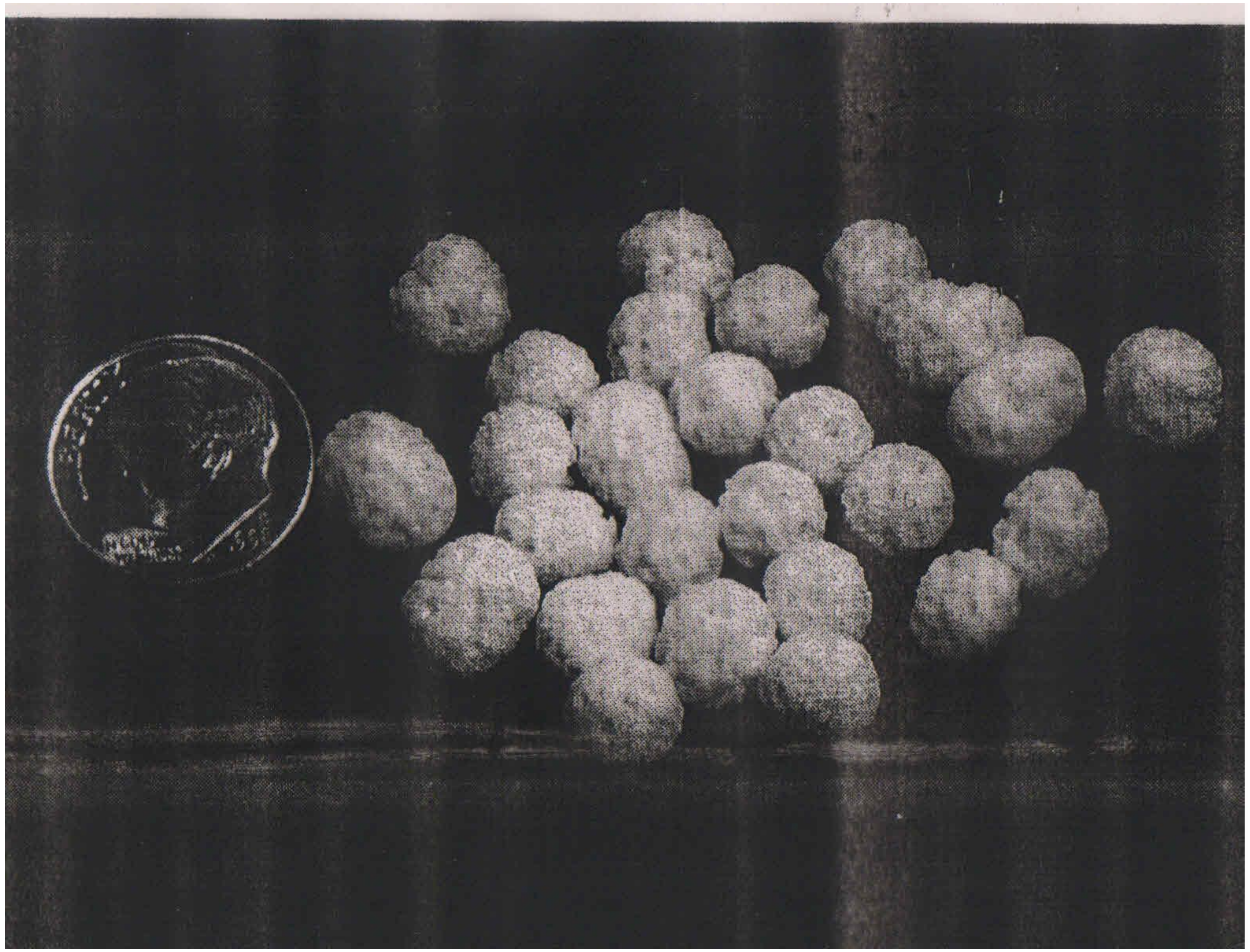
SiC Heating Elements. Other examples include $MoSi_2$, doped ZrO_2 and Graphite.



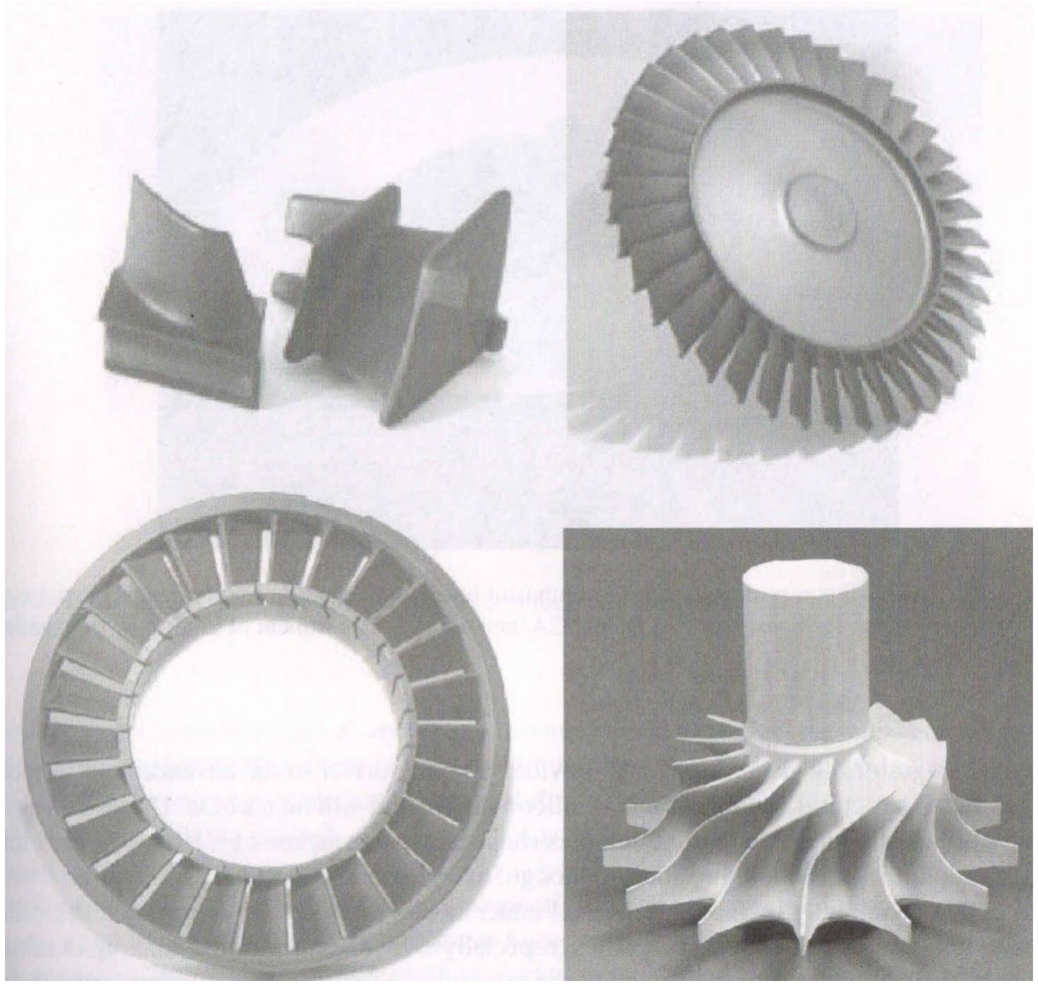
SiC tubular heat exchanger tubes with internal fins to increase heat transfer.



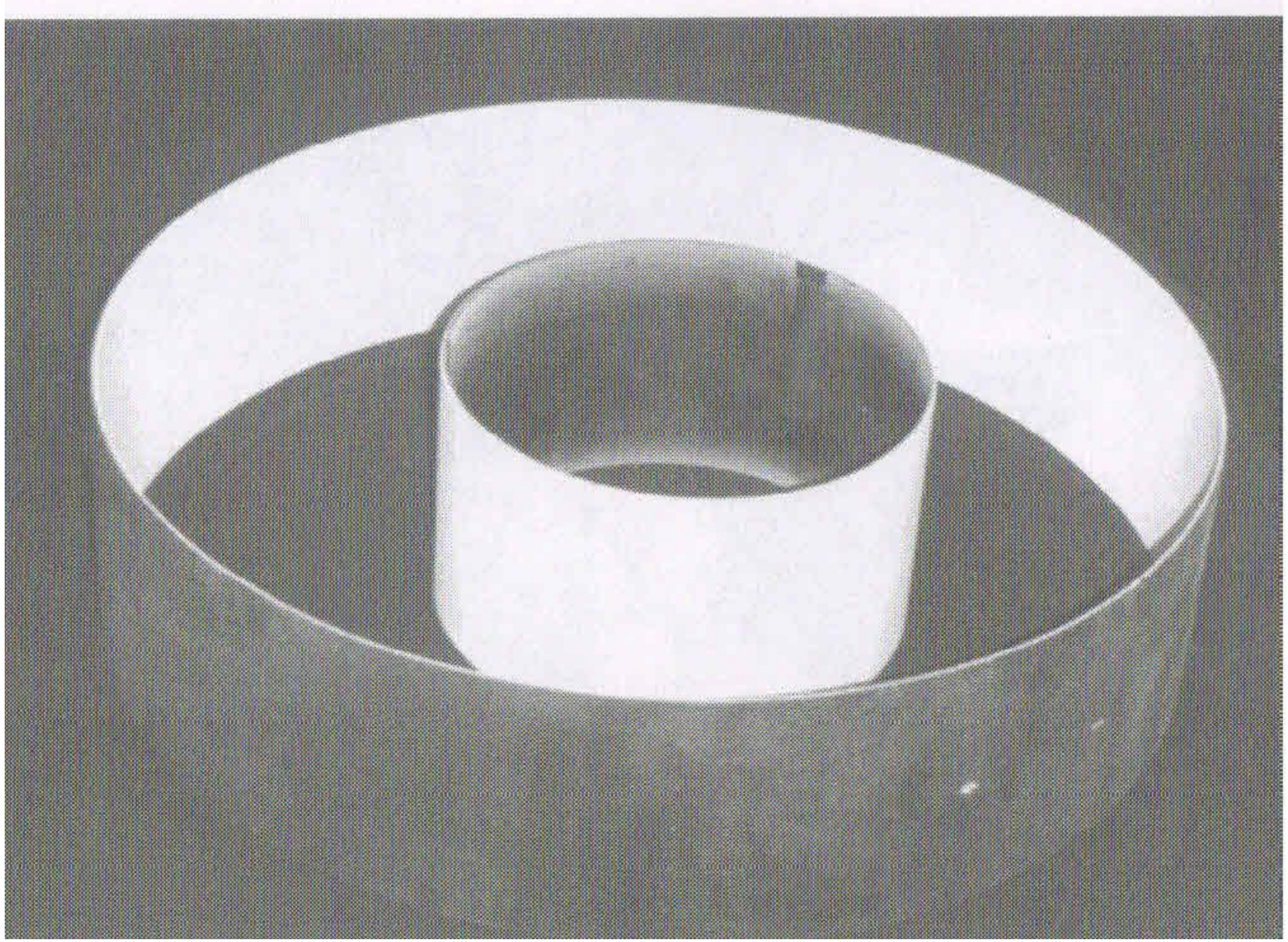
U-shaped SiC heat exchanger tubes fabricated in 1970.



Porous ceramic beads with high surface area to provide catalytic sites to accelerate chemical reactions in chemical processing such as in petrochemical cracking processes, in ammonia synthesis.



Examples of experimental Si_3N_4 and SiC gas turbine engine components that have been fabricated since 1975 for advanced gas turbine engine development program



SiC-SiC composite gas turbine combustor liners with an oxide environmental barrier coating.

Ceramics function as wear resistance

Wear resistance is the second major function of ceramics.

- **Ceramics have great potential as a means of reducing wear and friction.**

- **The potential advantages of ceramics over metals and polymers include high hardness, chemical stability, ability to be ground with very good surface finish with high dimensional tolerances, strength retention over a broad temperature range and low cost.**

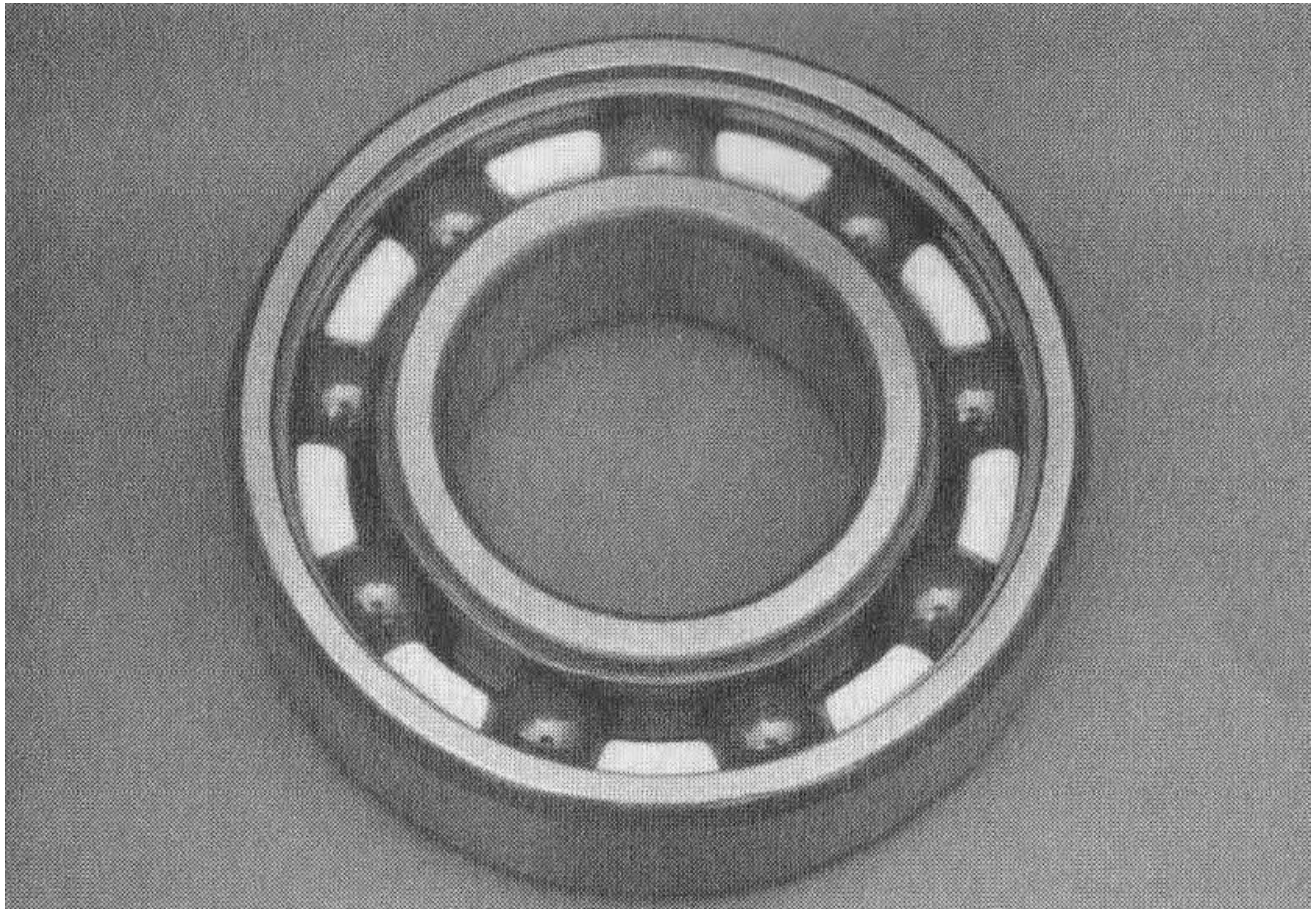
Wear resistant ceramics

Examples: **Al_2O_3, ZrO_2, WC, SiC, Si_3N_4**

APPLICATIONS

- **Seals (tip seals, face seals, piston rings, ball and seals, etc);**

- **Sand blast nozzles, wear pads, dies, pipes, liners, media for grinding & deburring, thread guides for textile industry, components for paper making, and a variety of protective and wear resistant coatings.**



High performance ceramic ball bearings and races are made from titanium and carbon nitride feedstocks through PM processing

Ceramics in shaping and finishing operations

EXAMPLES

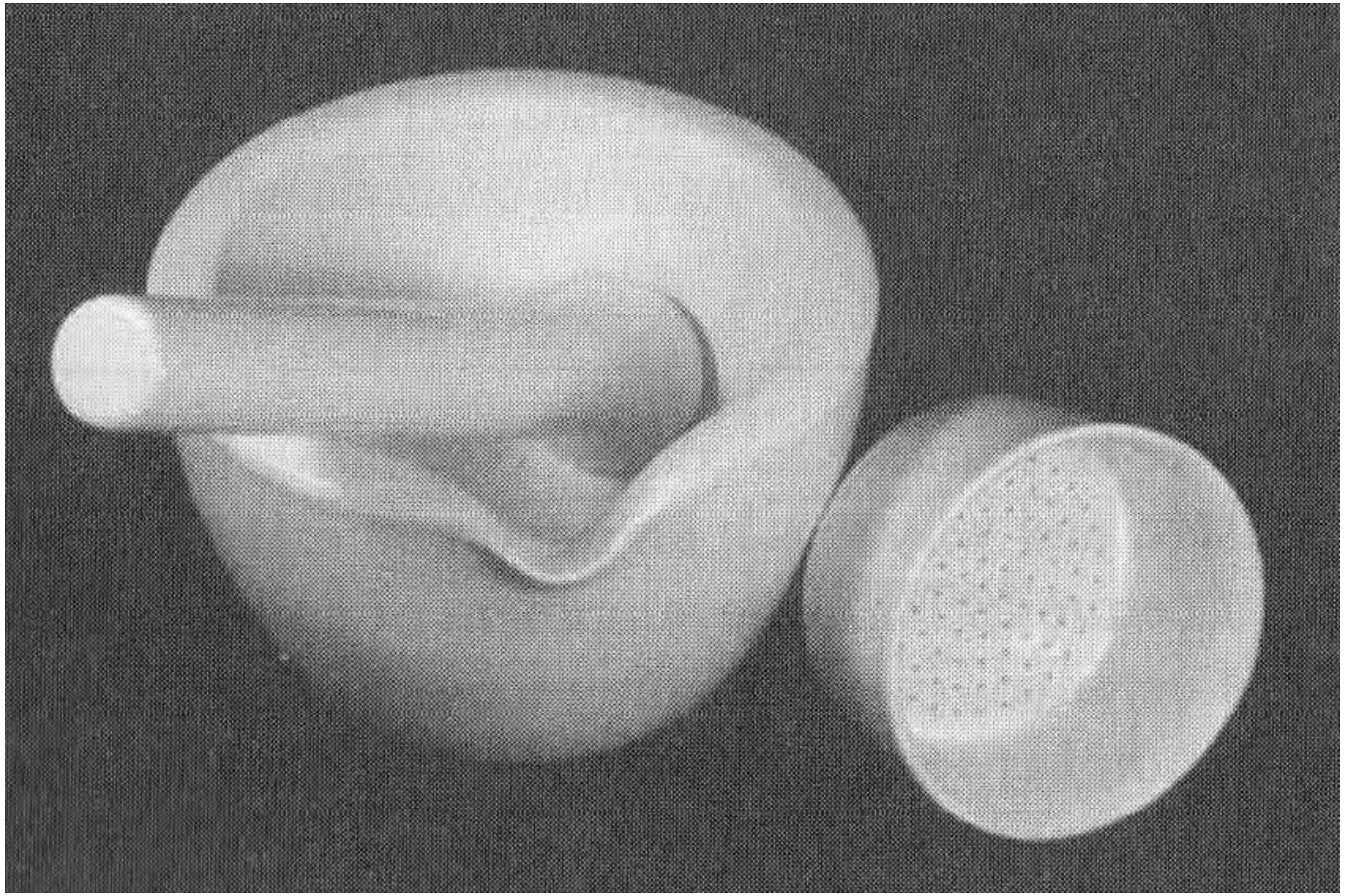
- Al_2O_3
- TiC
- TiN
- Sintered BN, B_4C
- Cermet (WC+Co)
- Diamond
- Si_3N_4

Applications in shaping and finishing

- **Traditional: Grinding wheels, sandpaper (and other coated abrasive products), sand blasting and peening.**

- **Less traditional applications include: cutting tool inserts, lasers, ultrasonic machining and water jet machining.**

Ceramic cutting tools have resulted in dramatic increases in the rate of turning operations for metal roughing and finishing operations.



Alumina laboratory ware (Pestle-mortar and Filter) meeting wear resistance and corrosion resistance.

Ceramics as Lasers

Lasers are a key element of most Laser systems (Rubby laser, Nd-glass laser, Nd-YAG laser) used for cutting, drilling, melting and welding operations.

Ceramics as Sensors and Actuators

- Ceramic sensors and actuators are the categories which are rapidly gaining importance in industry, especially as automation and robotics, become more widely used.

Ceramics Sensors

Ceramic sensors are used in industry for detecting the presence and approximate concentration of a wide variety of

- gases;
- rapid quantitative analysis of oxygen content of molten metals and automobile exhaust gases;
- dew point measurement; for combustible gas alarms; for temperature measurement and control;
- measurement of very small deflections;
- nondestructive inspection; and
- ultra precision position control.

Ceramics Actuators

Ceramic actuators based on piezoelectric and electrostrictive properties are used in

- **high precision machinery, especially for manufacturing of optical devices and cameras,**
- **ultrasonic generators for cleaners,**
- **ultrasonic machining, and**
- **ultrasonic non-destructive testing.**
- **$BaTiO_3$, and Lead zirconate titanate are widely used ultrasonic crystals.**

Ceramics for Optical functions

Optical characteristics make ceramics are useful as phosphors, lasers, fibres, lenses, and electromagnetic windows.

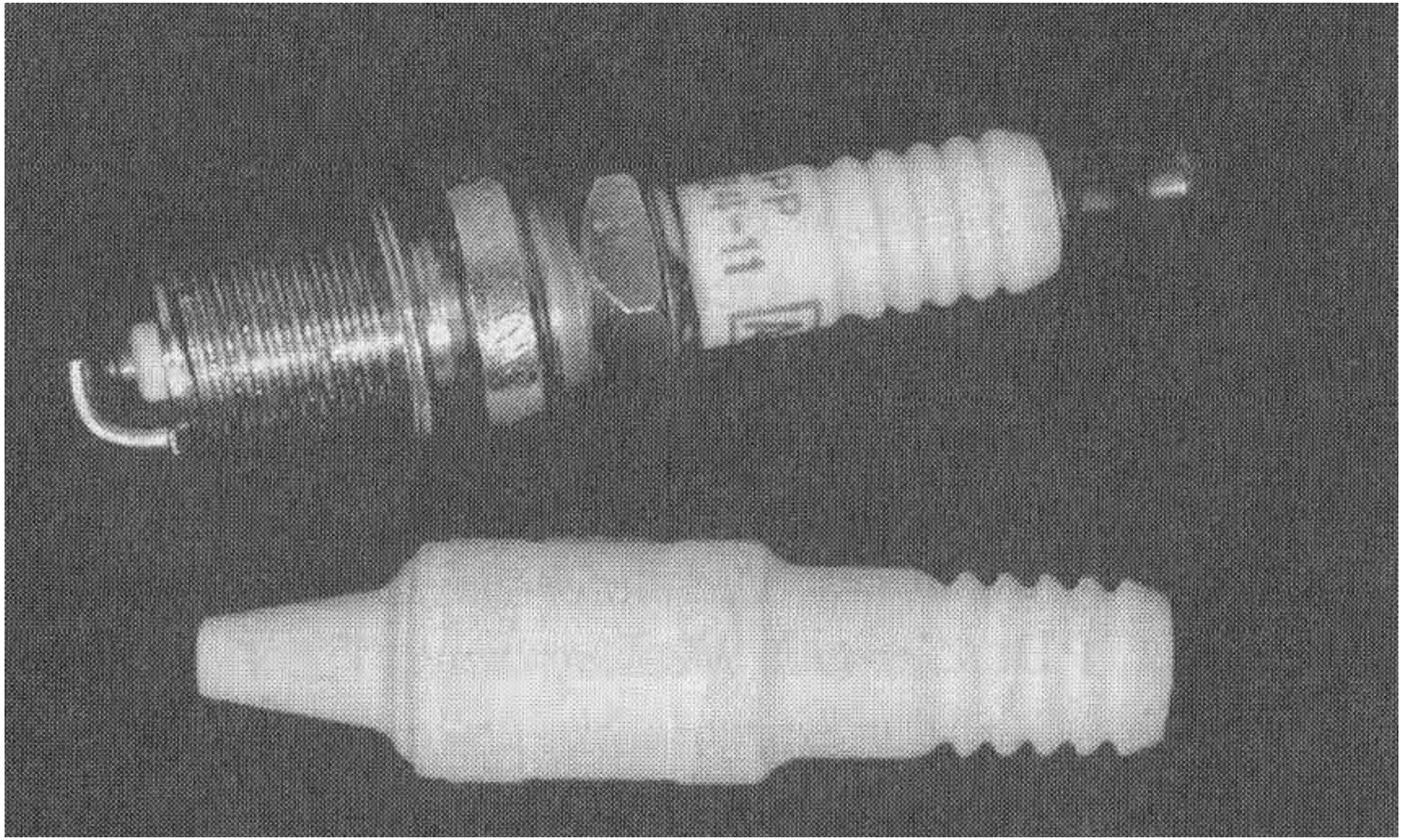
- A phosphor is a material which emits light when exposed to an energy source such as a beam of electrons. Ceramic phosphors produce the multicolour image on a TV/oscilloscope screen.

- Another ceramic phosphor produces the cool light in a common household fluorescent lamp.

Electrical function of Ceramics

- Electrical function of ceramics may be that of an insulator (Al_2O_3, MgO), heating element (SiC, $MoSi_2$), ignitor (SiC), dielectric ($BaTiO_3$, mica), piezoelectric ($BaTiO_3$, lead zirconate titanate), substrate (high purity Al_2O_3), oxygen sensor (ZrO_2), semi-conducting device, etc.

- Piezoelectric ceramics are used in manufacturing as ultrasonic generators for cleaners, as probe in non-destructive testing, ultrasonic machining, etc.



Alumina-based electrical insulator for an automotive spark plug plus an assembled spark plug fabricated by dry bag isotactic pressing. Spark plug insulator has to withstand high temperature pulses and severe thermal shock as well as chemical corrosion.

Ceramics for Magnetic Functions

Cubic ferrites and hexa-ferrites are used to perform Magnetic functions.

Hexagonal ferrites (especially Ba, Sr, and Pb hexaferrites) are frequently used for permanent magnets because of their high magnetization, compact size and low cost. Magnetic tape used for information recording and numerically controlled machines consists of fine dispersion of particles of a magnetic ceramic in flexible polymers tape. The magnetic ceramic particles perform the memory function.

Ceramics for Magnetic Functions (Contd…)

- Soft ceramic magnetic materials such as Mg-Zn and Ni-Zn spinel ferrites have high magnetic permeability and low loss and are suited for transformer and inductor applications.

Traditional vs. Advanced Ceramics

- Traditional ceramics are generally naturally occurring minerals with minimum preparation, i.e. with minimum purification and grinding. Therefore their cost is low.

- Raw materials include clay (china clay, fire clay and ball clay) , feldspar and flint (silica sand or quartz)

- These ceramics are inferior to advanced or technical ceramics.

Traditional vs. Advanced Ceramics

Advanced ceramics are prepared before processing to shaped products.

- Preparation involves purification of naturally occurring mineral, their grinding to proper size particles and their distribution and even addition of dopants or sintering aids to some ceramics.
- Sometimes advanced ceramics are prepared chemically.

Traditional vs. Advanced Ceramics

Why purification?

- The amount and type of impurities present in naturally occurring minerals greatly affect the final properties of the ceramic product. Some impurities are more deleterious than others, e.g. K or Na in electronic ceramic grade alumina, and transition metal ion in Zirconia deteriorate their functional properties.

Traditional vs. Advanced Ceramics

Particle size and distribution

- Small particle size (< 1μ) is very common with technical ceramic raw material. Even narrow size distribution is preferred to produce a dense product.

Small particles size means:

- Large surface area, therefore easy to sinter due to large reactivity
- Voids produced will be smaller
- In fact, microvoid incontrolled amount are sometimes desirable

Traditional vs. Advanced Ceramics

Use of Dopants and their distribution (homogeneous)

- Sometimes, impurities are added in controlled amount to a given ceramic in order to achieve some desirable property.

For example,

- small amount of MgO is added to alumina.
- CaO, MgO, or CeO_2, Y_2O_3 added to ZrO_2 to suppress phase transformation.
- small amount of La is added to $BaTiO_3$ to induce electrical conductivity

Traditional vs. Advanced Ceramics

- 2-3% of MgO, Al_2O_3 or Y_2O_3 is added as sintering aid to Si_3N_4 (metal oxide combines with SiO_2 film on the surface of Si_3N_4 and forms low melting point oxynitride glass which aids liquid phase sintering)

- The above impurities are added in small amounts at controlled rate and are commonly called as dopants similar to that done in intrinsic semiconductors.

- However a uniform dispersion/distribution poses severe mixing problems.

Traditional ceramics

- Composed of three ingredients: Clay, flint (silica), feldspar (potash feldspar $K_2O.Al_2O_3.6SiO_2$).

Functions of **Clay:**

- It constitutes the main body material of the traditional ceramics. Its fine particle size imparts good workability or plasticity required for forming of material to a particular shape before firing or sintering. Fine particle size also results is fine size pores.

- Clay also imparts strength to the ceramic body after firing. For a difficult forming technique, clay content of the ceramic must be high.

Traditional ceramics

Flint (silica)

- It has high melting point and therefore provides refractoriness to the ceramic product. It is unreactive at low temperature but forms a high viscosity liquid phase at high temperature.

Traditional ceramics

Feldspar (Potash feldspar $K_2O.Al_2O_3.6SiO_2$)

- It has low melting point and acts as a flux. It makes a glass (viscous liquid) when the ceramic mix is fired and bonds the refractory components together. Thus aids in vitrification.
- As the amount of feldspar is icreased, the amount of liquid forms at the eutectic temperature is increased, vitrification occurs at a lower temperature. A higher liquid content leads to greater vitrification and higher translucency.
- As feldspar is replaced by clay, higher temperatures are required for vitrification and firing process becomes difficult and expensive. However, forming process becomes easier and the mechanical and electrical properties of the resultant product would be superior.

Traditional ceramics

Examples of Traditional ceramics

- Hard porcelain
- Electrical insulators
- Sanitary ware
- Semi-vitreous whiteware
- Bone china
- Hotel china
- Dental porcelain

ADVANCED CERAMICS

- **Characteristics**

- High melting temperatures
- High hardness and stiffness
- High hot strength
- High compressive strength
- High wear and corrosion resistance
- Low density

ADVANCED CERAMICS

- Good dielectric properties
- Good thermal and electrical insulating properties
- Semiconductor and Ion conductor properties
- Magnetic properties
- Biocompatibility
- Improved toughness

ADVANCED CERAMICS

Oxide ceramics

Single metal oxides:

- Al_2O_3, BeO, ZrO_2, MgO, ThO_2, TiO_2, etc.

Mixed Oxides:

- Cordierite ($2MgO.5SiO_2.2Al_2O_3$)
- Steatite ($MgO.4SiO_2$)
- Forsterite ($2MgO.5SiO_2$)

Oxide Ceramics

- As a class, ceramic oxides are low in cost, except BeO and ThO_2
- They are produced by slip casting, pressing or extrusion and then fired at about 1800 °C
- However, they are more difficult to fabricate than ono-oxide ceramics with high density body and minimum distortion coupled with high dimensional stability.

Properties of Oxide Ceramics

Property	Alumina	Beryllia	Magnesia	Zirconia	Thoria
M.P., °C	2050	2570	2800	2677	3300
E, GPa	393	310 - 375	225	205	138
T.S., MPa	124 - 207	90 - 133	138	455, PSZ	52
C.S., MPa	1620 – 2585	1500	827	1861 – 2965	1378
Hardness (Knoop)	3000	1300	700	1100	700
Max. Service Temp., °C	1950	2400	2400	2500	2700

Properties of Oxide Ceramics

Property	Alumina	Beryllia	Magnesia	Zirconia	Thoria
Thermal Conductivity, W/mK	16 - 39	230 - 290	37	~ 3.0 (PSZ)	10
CTE, x10-6/°C	8.0	8.8	14.0	7.5	9.0

Alumina

Characteristics

- Most widely used oxide ceramic
- Hardest among oxide ceramic, possesses high tensile and compressive strength
- Outstanding electrical resistivity (~ 10^{15} Ω cm)
- Low α_{th} (~ 8 x 10^{-6}/°C)

Alumina

- Intermediate K_{th} (29-39 W/mK at R.T.)
- Density varies over wide range depending on impurity content and types, and method of fabrication.
- Commonly doped with MgO (0.25 to 0.5 wt%), pressed and sintered
- Good chemical resistance against air, water vapour, and suplphurous atmoshphere

Alumina

Applications

- High value added alumina parts are produced from ultrafine synthetic raw materials to control microstructure.
- Alumina with > 72% purity is characterized as high alumina bodies.
- Alumina with > 99% purity is characterized as high purity alumina and find applications in electroceramics & lamp envelopes; various wear components like pump seals, bearings, nozzles, thread guides, dental & prosthetic implants, textile guides, pump plungers, chute lining, discharge orifice and dies.

Alumina

- High strength substrates in electronic industries (as in packaging) and ZrO_2 dispersed high purity (99.7%) alumina find uses for body implants and cutting tools.

- Due to high hardness, alumina is also used as an abrasive

- As insulator in spark plug

- Specially doped alumina is used as a LASER (Ruby laser)

- As refractory
- As crucible material to hold metal

Alumina

- To operate at high temperatures where good strength is required. For instance, α-alumina is used for structural applications.
- High purity alumina is often used in production of translucent envelopes for Na-vapour lamps.

Magnesia (MgO)

Characteristics and Applications

Magnesia is a good electrical insulating material at elevated temperature. Because of this property, magnesia is used as (i) high temperature insulator encircling the heating elements used in electrical stoves and ovens and (ii) thermocouple sheath material to separate thermocouple wires and to protect them from direct exposure to high temperature environment.

Magnesia (MgO)

- Because of its high melting temperature, MgO is used as important refractory material in steel plant furnaces.

- Pure, dense material with ~ zero porosity is optically transparent and is useful candidate material for infrared transmitter in various systems such as radomes used as windows for electronic guidance and detection equipment on missiles, aircrafts and spacecrafts.

Magnesia (MgO)

- However, MgO is not as widely used as alumina, especially for structural use because of its lower strength, higher α_{th}, and therefore poor thermal shock resistance.

- Though superior in resistance against oxidation over alumina, MgO is less stable in contact with most metals at above 1700°C in reducing atmosphere or vacuum

- MgO has high vapour pressure > 1500 °C, therefore in practice it is used below this temperature.

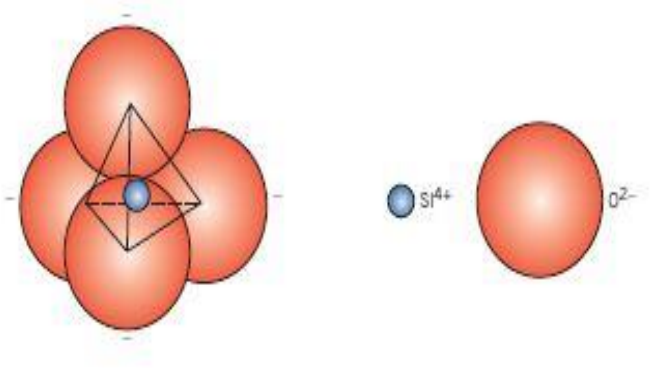
Silicates

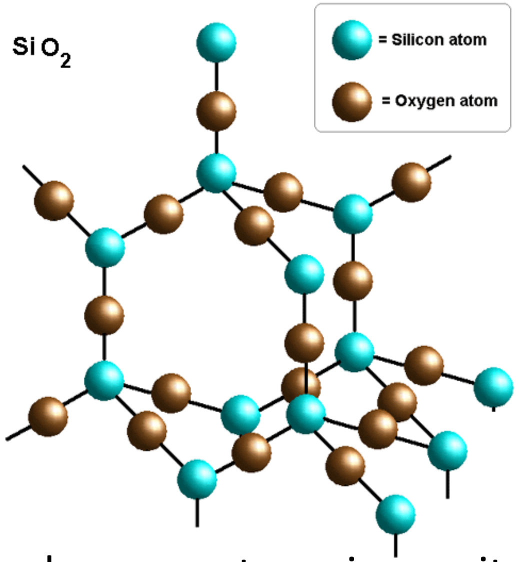
Each atom of silicon is bonded to four oxygen atoms which are situated at the corners of the tetrahedron and Si atom is positioned at the centre

Material is electrically neutral and all atoms have stable electronic structures

If these tetrahedra are arrayed in a regular and ordered manner, a crystalline structure is formed there are 3 primary polymorphic crystalline form of silica

1. Quartz
2. Critobalite
3. Tridymite

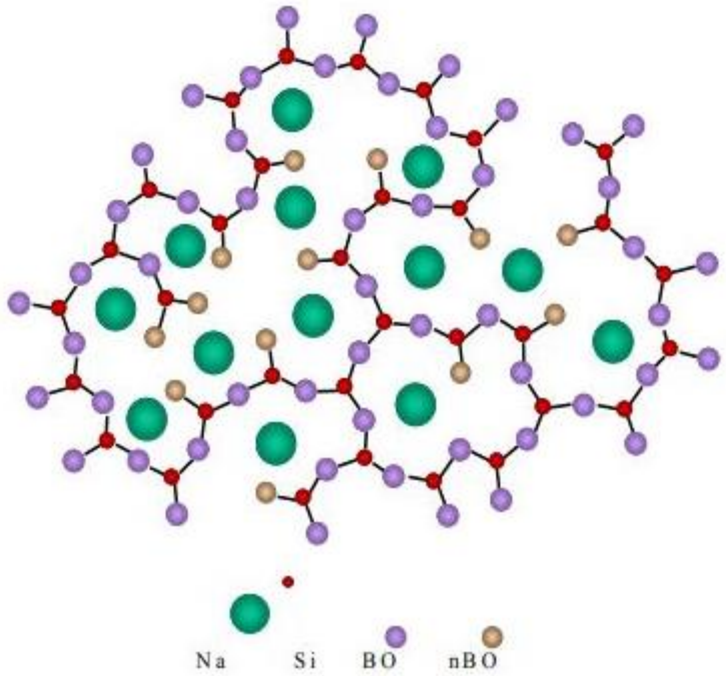




The arrangement of silicon and oxygen atoms in a unit cell of cristobalite, a polymorph of SiO_2

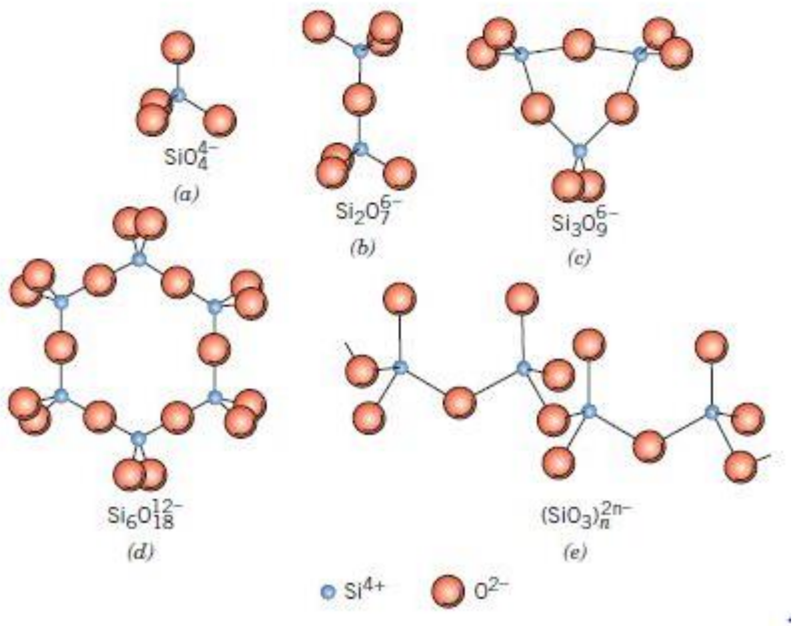
Silica glass

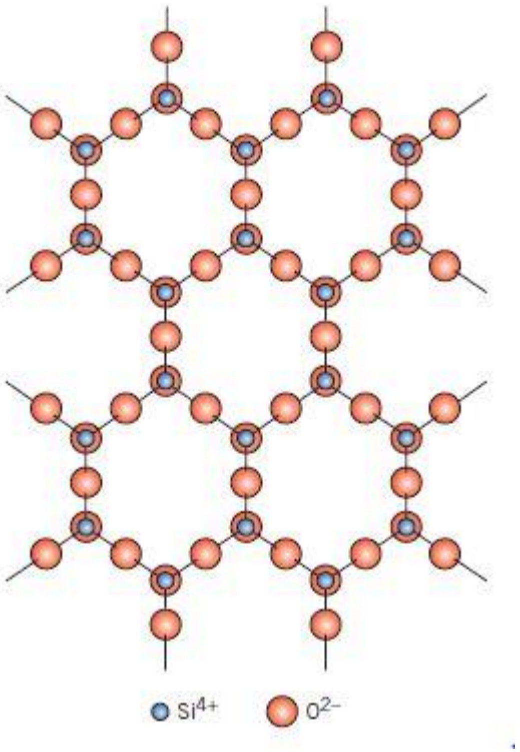
Silica can also be made to exist as a non crystalline solid or glass, having a high degree of atomic randomness, which is characteristic of the liquid, such a material is called fused silica or vitreous silica.

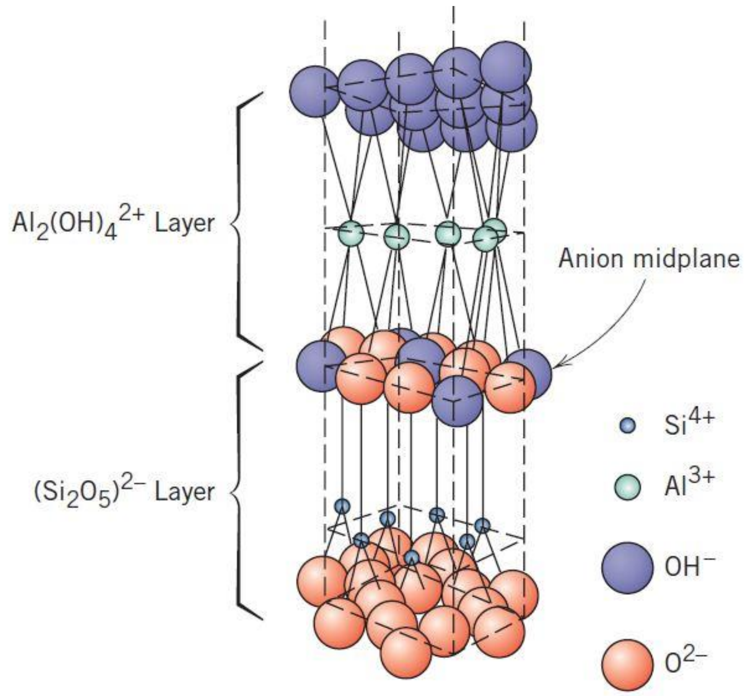


- The common inorganic glasses that are used for containers, windows, and so on are silica glass to which have been added other oxides such as CaO and Na_2O.
- These oxides are known as network modifiers generally they modify the network rather than form polyhedral
- Still other oxides TiO2 and Al2O3 are called as intermediates they do not form network but they substitute for Si and become part of it and stabilize the network
- Practical perspective these network modifiers and intermediate lowers the melting point and viscosity of a glass and makes it easier to form a lower temp

Simple silicate and layered structure



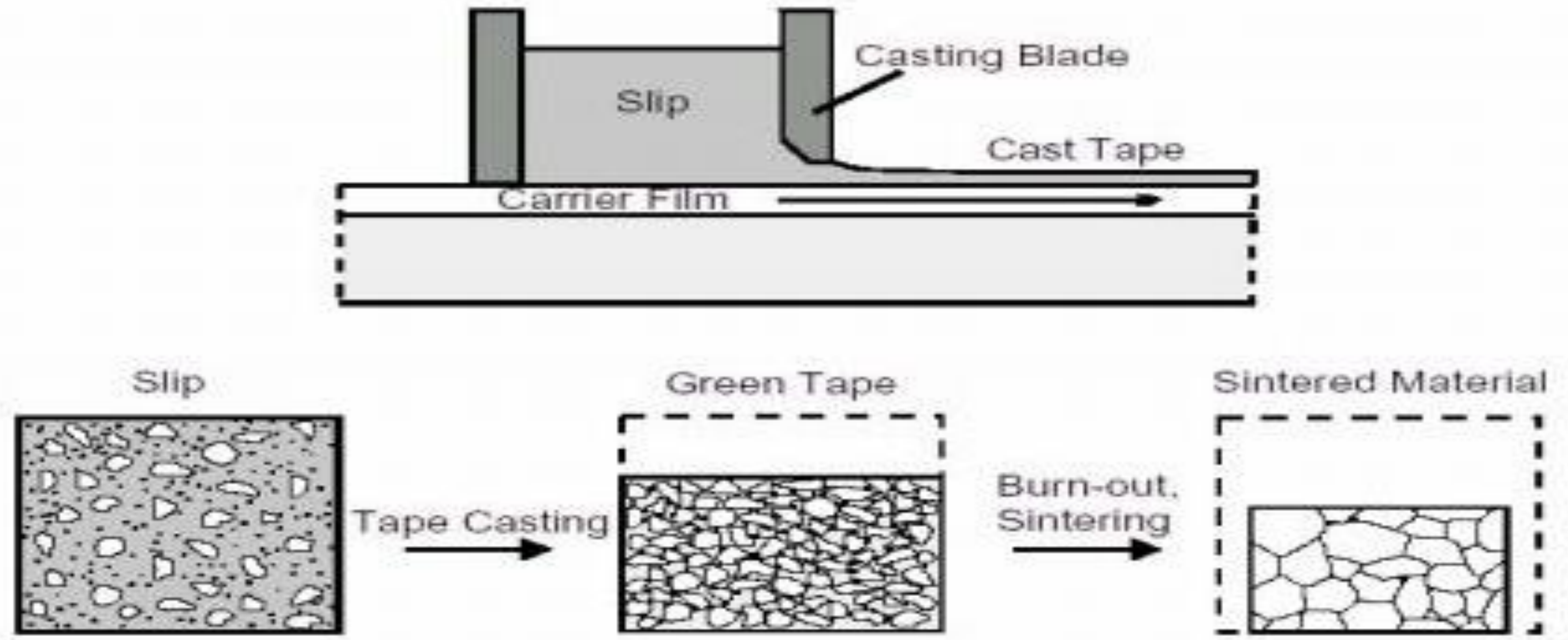




Manufacturing Techniques

Tape Casting

Thin sheets of a flexible tape are produced by casting. These sheets are prepared from slips. This slip contains a suspension of ceramics particles in an organic liquid that also contains binders and plasticizers to enhance strength and flexibility to cast.



The top schematic drawing illustrates tape casting. The bottom sketch shows the different stages during the processing: the slip consisting of water, ceramic particles and binder; the cast, dried green sheet; and, finally, the microstructure of the sintered material.

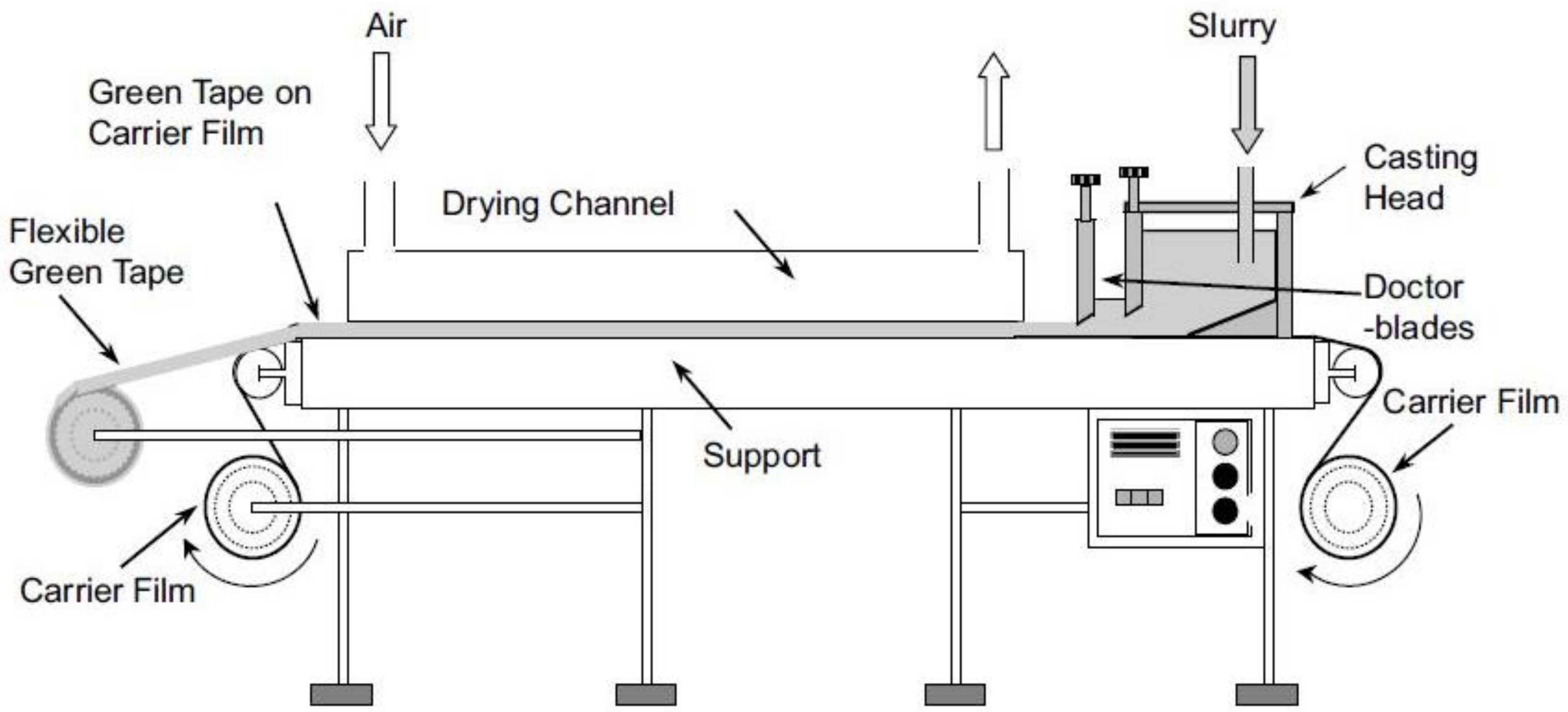
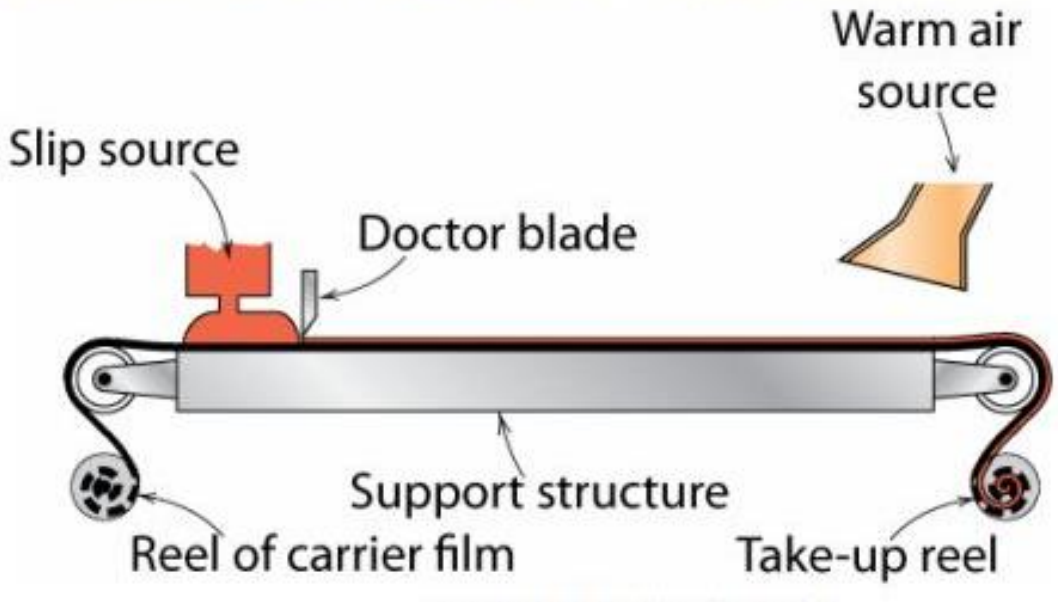


Fig. Schematic diagram of a tape casting machine

Tape Casting

- thin sheets of green ceramic cast as flexible tape
- used for integrated circuits and capacitors
- cast from liquid slip (ceramic + organic solvent)



Tape Casting

Thin sheets of a tape are produced.

Start from a slip (ceramic particles + organic liquid containing binders and plasticisers)

Thank you